

ANUPREET SINGH

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EDUCATION

University of Maryland, Baltimore County

(Master of Science in Computer Science, GPA: 3.8/4.0)

Baltimore, MD

Aug 2024 – May 2026

Indian Institute of Information Technology, Sonapat

(Bachelor of Technology in Computer Science and Engineering, GPA: 8.45/10)

Haryana, India

Aug 2019 – May 2023

SKILLS / RELEVANT COURSEWORK

Programming Languages: Python, JavaScript, C++, SQL

Tools and Frameworks: OpenCV, torchvision, PyTorch, Numpy, scikit-learn, LangChain, ONNX, Pinecone, React, PostgreSQL, PyTest, AWS, Docker, Kubernetes, GitHub Actions(CI/CD)

Areas of Interest and Coursework: AI & LLMs, Data Structures & Algorithms, Software Engineering (SDLC)

PROJECTS / ACADEMIC EXPERIENCE

When Pixels Talk Back | *Watermarking EMR Data to analyse impact on CV models*

- Fine-tuned **3 computer-vision models** (ResNet50, DenseNet169, MobileNetV2) for classification over COVID-19 lung CT scans via two-stage ImageNet transfer learning, to establish a baseline peaking at **93%** accuracy and **0.951 F1**.
- Benchmarked **3 watermarking algorithms** across all 3 models, showing the fragile scheme costs **0%** accuracy loss, versus ~1% for reversible embedding with a 300-bit payload and 11.2% for robust embedding.
- Embedded a 1024-bit blind fragile watermark into the Region-of-Non-Interest of CT scans via key-scrambled LSB embedding of a SHA-512 fingerprint, enabling content authentication with near-lossless quality at **SSIM 0.999**.

Nothing Lost in Translation | *Lexically constrained Sequence Generation*

- Built a Grid Beam Search decoder that guarantees positive/negative lexical constraints in any autoregressive sequence generation, demonstrated on English-to-Russian NMT with a **15-point BLEU** gain (25.45 → 40.59).
- Reconstructed a beam search decoder from scratch, outperforming **HuggingFace's** stock `.generate()` by **1.45 BLEU**.

Bloodborne TAG | *Text Adventure Game with natural language command parsing*

- Engineered an **AI powered command parser** using few-shot Mistral-7B for slot extraction and embedding similarity based intent matching, mapping free-form playtest commands to 14 deterministic actions at **96%** intent accuracy.
- Architected a data-driven game engine that loads a 41-room world and **60+** items/NPCs from JSON into in-memory session state, gating progression on generic key-item fields so new content ships with zero engine code changes.

Graduate Assistant | *UMBC - Data Structures & Software Engineering*

Jan 2025 – June 2026

- Mentored 10 student Scrum teams through sprint planning, reviews, and backlog refinement for their projects.
- Coordinated between 40 students and stakeholders, keeping all 10 projects on schedule amid evolving requirements.

INDUSTRY EXPERIENCE

Data AI Engineer | *Platinum Business Services (R&D Team) | Catonsville, MD*

Dec 2025 – Present

- Built a priority-ordered fallback in the **RAG chat agent** across 5 embedding models (OpenAI, Anthropic, Ollama), re-embedding and re-running low-citation queries against each successive model's vector index to exploit differences in embedding representation and surface SSP evidence missed by higher-priority indexes, raising evidence-backed control assessment **accuracy from 64% to 92%**.
- Diagnosed an out-of-memory failure in that fallback path, tracing it to freshly pulled 4–5 GB model weights lingering in the OS page cache, and fixed it with targeted `posix_fadvise` eviction, **eliminating model-load failures**.
- Developed backend services for an **AI-native** GRC platform that audits an SSP draft for compliance with NIST 800-53, chunking ~1,200 controls into a per-model vector index with linkage metadata tying each chunk to its parent control and sibling clauses, cutting a control-by-control review from **days of manual work to <10 minutes**.

Software Engineer (AI) Intern | *PMA | Tallahassee, FL*

Sep 2025 – Dec 2025

- Built an **AI-powered** PRD review platform in React and Django that compares each uploaded draft PDF against a reference PRD and emits schema-validated gap flags for a human-in-the-loop reviewer to confirm or dismiss, giving a cohort-wide matrix that replaces a read of every draft and cuts feedback-session prep from **8 to 2 hours**.
- Architected a fully serverless review pipeline on AWS (S3 + CloudFront-hosted React SPA, Django on Lambda, DynamoDB, and an SQS-triggered Lambda AI worker) that retries failed analyses off a dead-letter queue and logs per-report model API spend, sustaining 200+ submissions across 5 cohorts at ~\$3/month idle cost.